Tom Zhang

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Education

University of Waterloo Waterloo, ON
Bachelor of Computer Science (GPA: 3.9/4.0) *Sep 2022 – May 2026*

Experience

Software Engineer Intern May 2026 – Dec 2026
Cortico Health Technologies Inc. *Toronto, ON*

- Working on a **Django** healthcare platform for clinics, building robust **REST APIs** and containerizing full-stack applications using **Docker** and **Linux**, supporting **10k+ daily active users**.
- Engineered a real-time patient-clinic messaging system utilizing **Redis** and **WebSockets**, **cutting message latency under 100ms** and improving communication efficiency for **50+ clinics**.
- Worked on AI Inbox features for clinicians to summarize and categorize patient test results using **OpenAI API**, **saving clinicians an average of 2 hours daily** while ensuring strict data privacy by de-identifying patients.

Software Engineer Intern Sep 2025 – Dec 2025
Microart Services Inc. *Toronto, ON*

- Worked on a fullstack service for business data retrieval and visualization, **reduced response time by 19%** for **200+ users** by restructuring callbacks and debouncing user keystrokes using Python and Plotly
- Engineered an automated **ETL pipeline** using OpenCV and Tesseract to parse technical documents, **migrating 1,000+ legacy records** into a centralized MongoDB database for enhanced queryability

Software Developer May 2023 – June 2024
Centennial High School *Saskatoon, SK*

- Developed a Python-based automation tool using Openpyxl to transform raw class schedules into organized Excel documents, **saving several hours of manual work weekly for over 5 local schools**

Projects

Minesweeper Solver | *Python, Git, OpenCV, PyAutoGUI* May 2025 – June 2025

- Created a custom algorithm to autonomously solve Google Minesweeper on the browser, achieving an **accuracy of 99.9%** via a Backtracking Constraint Satisfaction (CSP) algorithm
- Utilized optimization techniques such as tree-pruning and meet-in-the-middle, improving the efficiency of the algorithm by 80%, and **consistently solving a 20x24 Minesweeper in under 20 seconds**
- Developed a multi-stage pipeline involving screen capture, image processing, an algorithm, and click signaling, ensuring high performance and throughput
- Designed the project with object-oriented principles and the MVC architecture in mind, ensuring high scalability

MarketSense Engine | *Python, FastAPI, Langchain, PostgreSQL, Docker, React, Tailwindcss* July 2025 – Aug 2025

- Engineered a high-performance Agent using FastAPI with **asynchronous programming** and non-blocking I/O, achieving sub-3-second response times for complex labor-market queries
- Architected a **scalable, multi-threaded** backend supporting **150+ concurrent connections** by leveraging PostgreSQL (pgvector) and optimized hybrid/multi-query retrieval pipelines
- Developed custom tool-calling interfaces (Google Search, Vector Search) and **containerized the infrastructure with Docker** to ensure seamless deployment and environment parity

Technical Skills

Languages: Python, JavaScript, HTML/CSS, SQL, C/C++, Java, Ruby, Go, Bash
Frameworks/Libraries: Node.js, Express, Django, React, Tailwind CSS, Flask, FastAPI, OpenCV, LangChain
Tools: Git, Linux, AWS, Docker, Jenkins, MySQL, PostgreSQL, MongoDB, ChromeDevTools, Cursor